# Train Your Own AI KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

**1. What should it learn?** Two things I could teach a camera AI to tell apart:

1. Answers vary. 2. _____

**2. How it works.** Match each Teachable Machine word.

| Teachable Machine | What it means |
|---|---|
| Class | one kind of thing: a label, like “rock” |
| Samples | the training pictures for a class |
| Train Model | the network fixes its weights |
| Preview | testing it on new pictures |

**3. Train it.**

| Class name | Number of samples | Did the preview get it right? |
|---|---|---|
| Rock | about 50 | |
| Paper | | |

**4. Fool it.** Change one thing at a time.

| What I changed | What the AI said | Fooled? (yes / no) |
|---|---|---|
| my other hand | Answers vary. | |
| very close, then far away | | |
| lights off or a bright window | | |
| my partner's hand | | |

Why did some tricks fool the AI? It never saw anything like them in its training pictures.

An AI is only as good as its training examples.

**5. Talk about it.** Would your AI work for everyone?

My training pictures were missing Answers vary. Other people's hands, other light, ...

**Go Further (Day 2).** Make it harder to fool.

| What I changed | What the AI said | Fooled? (yes / no) |
|---|---|---|
| a new background | Answers vary. | |
| a glove or a sock | | |

How could you make it harder to fool? Add more kinds of examples and train again.

After I trained again, the AI was Answers vary. Harder to fool.